A Scalable Algorithm for Bayesian Variable Selection (SAB) with Application to miRNA-mRNA Regulation in Cancer

Jin Wang, Feng Liang, Yuan Ji and Yitan Zhu

1 Introduction

1.1 Statistical Background

Consider the following canonical setup for a linear regression model: given a dependent variable Y and p independent predictor variables $X_1, \ldots, X_p$, we model Y as

$$Y = X_1\beta_1 + X_2\beta_2 + \cdots + X_p\beta_p + \epsilon, \tag{1}$$

where β_j are the unknown regression coefficients and the error term ϵ is assumed to follow a normal distribution. A long-standing problem in regression analysis is to identify variables that are truly (if the assumed model is correct) or approximately relevant to the response Y, i.e., variables whose coefficients β_j are non-zero in Equation (1).

Many variable selection methods are based on the framework of penalized likelihood. In the context of linear regression, the penalized likelihood usually takes the form

Jin Wang
Department of Statistics, University of Illinois at Urbana-Champaign, Champaign, IL, USA, e-mail: jinwangmls@gmail.com

Feng Liang
Department of Statistics, University of Illinois at Urbana-Champaign, Champaign, IL, USA, e-mail: liangf@illinois.edu

Yuan Ji
Department of Public Health Sciences, University of Chicago, Chicago, IL, e-mail: koaeraser@gmail.com

Yitan Zhu
Program of Computational Genomics & Medicine, Northshore University Health System, Evanston, IL, USA, e-mail: yzhu@northshore.org

Appendices

A Implementation of SAB

A.1 Stopping Criteria

SAB iteratively updates $(\mathbf{m}, \mathbf{D}, \phi_{1:p}, \theta, \sigma^2)$ to minimize the objective function (12) until convergence. A natural stopping criterion would be based on the change of the objection function. However, evaluating the objective function in each iteration requires some additional computation that is irrelevant to the updating procedure, such as computing the determinant of $\mathbf{D}$, a $p \times p$ matrix, which increases the overall computation cost. Since our primary interest is variable selection, what matters is the convergence of the variable inclusion probabilities ϕ_j. Therefore we monitor the entropy of the parameter $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_p)$:

$$\mathsf{Ent}(\boldsymbol{\phi}) = -\sum \phi_j \log \phi_j - \sum (1 - \phi_j) \log(1 - \phi_j), \tag{28}$$

and stop the iteration when the change of $\mathsf{Ent}(\boldsymbol{\phi})$ is below some pre-specified threshold.

A.2 Fast Computation of Matrix Inverse

In SAB, at each iteration, we need to compute the inverse of a $p \times p$ matrix $\mathbf{V}_t = (\mathbf{X}^T\mathbf{X} + \boldsymbol{\Delta}_t)^{-1}$, where

$$\boldsymbol{\Delta}_t = \mathrm{diag}\Big(\frac{1 - \phi_j^{(t)}}{v_0} + \frac{\phi_j^{(t)}}{v_1}\Big)_{j=1}^{p}$$

and $\phi_j^{(t)}$ denotes the value of ϕ_j updated by formula (15) at the t-th iteration.

When $p > n$, we can use the Sherman-Morrison-Woodbury formula to compute the inverse of an $n \times n$ matrix instead, namely,

$$(\mathbf{X}^T\mathbf{X} + \boldsymbol{\Delta}_t)^{-1} = \boldsymbol{\Delta}_t^{-1} - \boldsymbol{\Delta}_t^{-1}\mathbf{X}^T(\mathbf{I}_n + \mathbf{X}\boldsymbol{\Delta}_t^{-1}\mathbf{X}^T)^{-1}\mathbf{X}\boldsymbol{\Delta}_t^{-1}. \tag{29}$$

So at each iteration, the computation cost for inverting this matrix is of the order $O(\min(n, p)^3)$, which is still time-consuming when both n and p are large.

Next, we describe a sub-optimal updating rule to reduce the computation cost. Note

$$\mathbf{V}_{t+1} = (\mathbf{X}^T\mathbf{X} + \boldsymbol{\Delta}_{t+1})^{-1} = (\mathbf{V}_t^{-1} + \boldsymbol{\Delta}_{t+1} - \boldsymbol{\Delta}_t)^{-1}. \tag{30}$$

Suppose we only update those ϕ_j's whose change is greater than a threshold (e.g., 0.01). Let l be the number of ϕ_j's that are updated at iteration $(t+1)$. Then the matrix $\mathbf{\Delta}_{t+1} - \mathbf{\Delta}_t$ is a rank-l diagonal matrix with entries equal to zero except the l diagonal entries where $\phi_j^{t+1} \neq \phi_j^{(t)}$. So we can write

$$\mathbf{\Delta}_{t+1} - \mathbf{\Delta}_t = J_t G_l J_t^T$$

where G_l is an $l \times l$ diagonal matrix and J_t is a $p \times l$ matrix with entries $0/1$.

Applying the Woodbury formula, we have

$$(\mathbf{X}^T\mathbf{X} + \mathbf{\Delta}_{t+1})^{-1} = \mathbf{V}_t - \mathbf{V}_t J_t \left[G_l^{-1} + J_t^T \mathbf{V}_t J_t\right]^{-1} J_t^T \mathbf{V}_t. \tag{31}$$

That is, we only need to inverse an $l \times l$ matrix, so the computation cost is just $O(l^3)$.

To summarize, when the dimension n is small, we can use (29) to reduce the computation from $O(p^3)$ to $O(n^3)$. When both n and p are large, we can use the sub-optimal updating rule, i.e., only update ϕ_j's whose changes between two adjacent iterations are bigger than some threshold, then we can further reduce the computation to $O(l^3)$ where l is the number of ϕ_j's that are updated in the iteration.

B Proofs

B.1 Proof for Theorem 1

Recall

$$\begin{aligned}
\log \frac{\phi_{nj}}{1-\phi_{nj}} &= \frac{1}{2}\log\frac{v_0}{v_1} + \frac{m_{nj}^2 + D_{njj}}{2\hat{\sigma}_n^2}\left(\frac{1}{v_0} - \frac{1}{v_1}\right) + \log\frac{\hat{\theta}_n}{1-\hat{\theta}_n} \\
&= -\frac{1}{2}\log\frac{v_1}{v_0} + \left[\left(\frac{m_{nj}}{\sqrt{v_0}}\right)^2 + \frac{D_{njj}}{v_0}\right]\left(\frac{1 - v_0/v_1}{2\hat{\sigma}_n^2}\right) + \log\frac{\hat{\theta}_n}{1-\hat{\theta}_n},
\end{aligned}$$

where m_{nj} denotes the jth element of $\mathbf{m}_n$ and D_{njj} denotes the jth diagonal entry of matrix $\mathbf{D}_n$. The updating formulae for $\mathbf{m}_n$ and $\mathbf{D}_n$ are

$$\begin{aligned}
\mathbf{m}_n &= (\mathbf{X}_n^T\mathbf{X}_n + \Delta_n)^{-1}\mathbf{X}_n^T(\mathbf{X}_n\boldsymbol{\beta}^* + \mathbf{e}_n) \\
&= \boldsymbol{\beta}^* - \left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}\Delta_n\boldsymbol{\beta}^* + \left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}\mathbf{X}_n^t\mathbf{e}_n \\
&= \boldsymbol{\beta}^* - \mathbf{b}_n + \mathbf{W}_n,
\end{aligned} \tag{32}$$

$$\mathbf{D}_n = \hat{\sigma}_n^2\left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}, \quad \Delta_n = \mathrm{diag}\left(\frac{\phi_{nj}}{v_1} + \frac{1-\phi_{nj}}{v_0}\right).$$

In (32), we express $\mathbf{m}_n$ into three terms: the true coefficient $\boldsymbol{\beta}^*$, the bias term $\mathbf{b}_n$, and the projection of the error $\mathbf{W}_n$. Note that $\mathbf{W}_n$ no longer follows a Gaussian distribution since Δ_n is random depending on $\mathbf{e}_n$.

First, we list some key results which will be used in the proof.

- (R1) With condition (C4), we have $\hat{\sigma}_n^2 = 1$ and $|\log \hat{\theta}_n/(1-\hat{\theta}_n)|$ bounded by a constant.
- (R2) The diagonal entries of $\mathbf{D}_n$ can be bounded from above by its largest eigenvalue, which is further bounded above by $(\lambda_{n1} + 1/v_1)^{-1}$,

$$\frac{D_{njj}}{v_0} \leq \frac{1}{v_0}\frac{1}{\lambda_{n1} + 1/v_1} = O(n^{r_0 - \eta_1}).$$

- (R3) The bias term b_{nj} is bounded by

$$\begin{aligned}\max_j \frac{|b_{nj}|}{\sqrt{v_0}} &\leq \frac{\|\mathbf{b}_n\|_2}{\sqrt{v_0}} = \frac{1}{\sqrt{v_0}}\left\|\left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}\Delta_n\boldsymbol{\beta}^*\right\| \\ &\leq \frac{1}{\sqrt{v_0}}\frac{1/v_0}{\lambda_{n1} + 1/v_1}\|\boldsymbol{\beta}^*\| \\ &= O(n^{3r_0/2 - \eta_1 + \eta_2}),\end{aligned}$$

  which goes to 0 when $r_0 < \frac{2}{3}(\eta_1 - \eta_2)$.
- (R4) With probability tending to unity, the maximum of $\mathbf{W}_n/\sqrt{v_0}$ is bounded above by a constant. Although $\mathbf{W}_n$ is not a Gaussian random vector due to the dependence between Δ_n and $\mathbf{e}_n$, we can write it as an (random) affine transformation of a multivariate Gaussian random variable:

$$\begin{aligned}\mathbf{W}_n &= \left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}\mathbf{X}_n^T\mathbf{e}_n \\ &= \left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}\left(\mathbf{X}_n^T\mathbf{X}_n\right)\left(\mathbf{X}_n^T\mathbf{X}_n\right)^{-1}\mathbf{X}_n^T\mathbf{e}_n \\ &= \left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}\left(\mathbf{X}_n^T\mathbf{X}_n\right)\tilde{\mathbf{W}}_n,\end{aligned}$$

  where each $\tilde{W}_{nj}$ is a Gaussian random variable with mean zero and variance bounded by λ_{n1}^{-1}. Let $A = \left(\mathbf{X}_n^T\mathbf{X}_n + \Delta_n\right)^{-1}\left(\mathbf{X}_n^T\mathbf{X}_n\right)$. Using the result on $\|A\|_\infty$ (the infinity norm) and $\|A\|_2$ (the spectral norm), we have

$$\max_j |W_{nj}| \leq \|A\|_\infty \max_j |\tilde{W}_{nj}| \leq \sqrt{p}\|A\|_2 \max_j |\tilde{W}_{nj}|.$$

  Note that $\|A\|_2$ is equal to its largest singular value which is the same as the largest eigenvalue of $\left(\mathbf{I}_p + (\mathbf{X}_n^T\mathbf{X}_n)^{-1/2}\Delta_n(\mathbf{X}_n^T\mathbf{X}_n)^{-1/2}\right)^{-1}$ denoted as $(\mathbf{I}_p + L)^{-1}$. Since L is positive definite, any eigenvalue of $(\mathbf{I}_p + L)$ should be greater than 1. Therefore we have

$$\|A\|_2 = \|(\mathbf{I}_p + L)^{-1}\|_2 \leq 1.$$

  Note that a Gaussian random variable $Z \sim \mathsf{N}(0, \tau^2)$ has its tail probability bounded by

$$\mathbb{P}(|Z| > t) = \mathbb{P}(|Z|/\tau > t/\tau) \leq \sqrt{\frac{2}{\pi}}\frac{\tau}{t}e^{-\frac{1}{2}\frac{t^2}{\tau^2}},$$

for any positive t. Recall that $\tilde{W}_{nj}$ is a Gaussian random variable with variance upper bounded by λ_{n1}^{-1}. So we have that for any constant $c > 0$, there exists another constant $M > 0$ such that

$$\mathbb{P}\Big(\frac{\sqrt{p}|\tilde{W}_{nj}|}{\sqrt{v_0}} > c\Big) \leq O\left(\sqrt{\frac{p}{\lambda_{n1}v_0}}\exp\{-\frac{c^2}{2}\frac{\lambda_{n1}v_0}{p}\}\right).$$

Apply the Bonferonni's inequality, and the fact that $\|A\|_2 \leq 1$, for any $c > 0$,

$$\begin{aligned}\mathbb{P}\Big(\max_j \frac{|W_{nj}|}{\sqrt{v_0}} > c\Big) &\leq \mathbb{P}\Big(\max_j \frac{\sqrt{p}|\tilde{W}_{nj}|}{\sqrt{v_0}} > c\Big)\\ &\leq O\left(p\sqrt{\frac{p}{\lambda_{n1}v_0}}\exp\{-\frac{c^2}{2}\frac{\lambda_{n1}v_0}{p}\}\right)\\ &= o\big(\exp\{-Mn^{\eta_1-r_0-\alpha}\}\big), \end{aligned} \tag{33}$$

for some constant $M < c^2/2$, which goes to 0 when $r_0 \leq \eta_1 - \alpha$.

Next, we prove (22). For $\boldsymbol{\gamma}_j^* = 0$, we have $\beta_j^* = 0$ and by results (R2) and (R4),

$$\begin{aligned}\max_{j:\gamma_j^*=0}\Big(\frac{m_{nj}^2}{v_0} + \frac{D_{njj}}{v_0}\Big) &\leq \Big(\max_j \frac{|b_{nj}|}{\sqrt{v_0}} + \max_j \frac{|W_{nj}|}{\sqrt{v_0}}\Big)^2 + \max_j \frac{D_{njj}}{v_0}\\ &\leq O(n^{3r_0-2\eta_1+2\eta_2}) + \left(\max_j \frac{|W_{nj}|}{\sqrt{v_0}}\right)^2 + O(n^{r_0-\eta_1})\end{aligned}$$

Under the condition $r_0 < 2(\eta_1 - \eta_2)/3$, both the squared bias term and the variance term go to zero, so there exists a constant $C > 0$ that for large enough n,

$$\mathbb{P}\Big(\max_{j:\gamma_j^*=0}\Big(\frac{m_{nj}^2}{v_0} + \frac{D_{njj}}{v_0}\Big) > c^2\Big) \leq \mathbb{P}\Big(\max_j \frac{|W_{nj}|}{\sqrt{v_0}} > c\Big) = o\big(\exp\{-Mn^{\eta_1-r_0-\alpha}\}\big)$$

By results (R1), with probability $o(e^{-Mn^{\eta_1-r_0-\alpha}})$,

$$\max_{j:\boldsymbol{\gamma}_j^*=0} \log\frac{\phi_{nj}}{1-\phi_{nj}} = -\frac{1}{2}\log\frac{v_1}{v_0} + O(1) = -O(\log n) < -C_n.$$

For $\boldsymbol{\gamma}_j^* = 1$, by results (R2-R4) we have

$$\begin{aligned}\min_{j:\gamma_j^*=1}\big(\frac{m_{nj}^2}{v_0} + \frac{D_{njj}}{v_0}\big) &\geq \Big(\min_{j:\gamma_j^*=1}\frac{|\beta_j^*|}{\sqrt{v_0}} - \max_j \frac{|b_{nj}|}{\sqrt{v_0}} - \max_j \frac{|W_{nj}|}{\sqrt{v_0}}\Big)^2\\ &\geq O(n^{r_0-2\eta_3}) - O(n^{3r_0-2\eta_1+2\eta_2}) - \left(\max_j \frac{|W_{nj}|}{\sqrt{v_0}}\right)^2.\end{aligned}$$

Under the condition $r_0 > 2\eta_3$, the dominate order of the left hand side should be $O(n^{r_0-2\eta_3})$ which is greater than $-\frac{1}{2}\log\frac{v_1}{v_0} = -O(\log n)$, with probability $o(e^{-Mn^{\eta_1-r_0-\alpha}})$. Therefore

$$\max_{j:\gamma_j^*=1} \log\frac{\phi_{nj}}{1-\phi_{nj}} = O(n^{r_0-2\eta_3}) > C_n.$$

Combining the two inequalities, (22) holds for $C_n = o(\log n)$, and we have

$$\mathbb{P}(\hat{S}_n = S_n^*) = 1 - o(p\exp\{-n^{\eta_1-r_0}\}) \to 1.$$

B.2 Proof for Theorem 1 when $p \gg n$

The proof is similar to the previous Section B.1 with the following changes. First note that the SAB algorithm starts with all variables being excluded; so we have $\Delta_n = \frac{1}{v_0}\mathbf{I}$ being a constant matrix.

- (R2) The new bound for the diagonal entries of $\mathbf{D}_n$ is given by

$$\frac{D_{njj}}{v_0} \le \frac{1}{v_0}\frac{1}{0+1/v_0} = 1.$$

- (R3) The bias term b_{nj} is now bounded by

$$\begin{aligned}
\max_j \frac{|b_{nj}|}{\sqrt{v_0}} &\le \frac{\|\mathbf{b}_n\|_\infty}{\sqrt{v_0}} = \frac{1}{\sqrt{v_0}}\left\|\left(\mathbf{X}_n^T\mathbf{X}_n + \frac{1}{v_0}\mathbf{I}\right)^{-1}\frac{1}{v_0}\mathbf{I}\boldsymbol{\beta}^*\right\|_\infty \\
&= \frac{1}{\sqrt{v_0}}\left\|\left(v_0\mathbf{X}_n^T\mathbf{X}_n + \mathbf{I}\right)^{-1}\mathbf{P}\boldsymbol{\beta}^*\right\|_\infty + \frac{1}{\sqrt{v_0}}\left\|\left(v_0\mathbf{X}_n^T\mathbf{X}_n + \mathbf{I}\right)^{-1}(\boldsymbol{\beta}^* - \mathbf{P}\boldsymbol{\beta}^*)\right\|_\infty \\
&\le \frac{1}{\sqrt{v_0}}\frac{1/v_0}{\lambda_{n1}+1/v_1}\|\boldsymbol{\beta}^*\| + \frac{1}{\sqrt{v_0}}o(n^{-r_n}\log n) \\
&= O(n^{3r_0/2-\eta_1+\eta_2}) + O(n^{-r_0/2}\log n)
\end{aligned}$$

  which goes to 0 when $r_0 < \frac{2}{3}(\eta_1 - \eta_2)$.
- (R4) Note that $\mathbf{W}_n = (\mathbf{X}_n^T\mathbf{X}_n + \frac{1}{v_0}\mathbf{I})^{-1}\mathbf{X}_n^T\mathbf{e}_n$ now is a Gaussian random vector with maximum variance upper bounded by $\sigma^2\lambda_{n1}^{-1}$. So we do not need $\tilde{\mathbf{W}}_n$. Using the union bound, we can show that

$$\begin{aligned}
\mathbb{P}\left(\max_j \frac{|W_{nj}|}{\sqrt{v_0}} > c\right) &\le O\left(p\sqrt{\frac{1}{\lambda_{n1}v_0}}\exp\{-\frac{c^2\lambda_{n1}v_0}{2}\}\right) \\
&= O\left(e^{-Mn^{\eta_1-r_0}+\log p}\right),
\end{aligned}$$

  which goes to zero when $\log p = o(n^{\eta_1-r_0})$.

Combining all the results, similar to the proof in Appendix B.1, we can show Theorem 1 when $\log p = o(n^{\eta_1 - r_0})$. Consequently, estimation consistency and Bayesian consistency can be achieved by the estimator returned by our SAB algorithm.